

Nantong university

Course Design

Design Topic: Engineering Structure (Single-story factory building)

Course Title: Reinforced Concrete Structures

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Class: Civil Engineering

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1 Design conditions and materials

The metalworking workshop of a machinery factory is a single-span industrial building with a span of 18m and a column spacing of 6m . The total length of the workshop is 60m. The span is equipped with two 200/50kN cranes , with a crane working class of A5 . The indoor floor elevation is ±0.000, the outdoor floor elevation is -0.15m , the distance from the indoor floor to the top of the foundation is 0.5m , the corbel top elevation is 8.7m, the crane rail top height is 9.600m , the column top elevation is 12.3m , the eaves elevation is 13.5m , and the roof elevation is 14.25 m. The lower window opening of the steel windows between the columns is 3.6m × 4.8m ; the upper window opening is 3.6m × 1.8m . The window sill elevations are 1m and 9.8m respectively . The building features a rolled waterproof roof, 370mm thick double-sided exposed brick walls, and a plain concrete floor. A cross-section of the factory building is shown in Figure 1. The factory is located...

The basic wind pressure at the location is 0.40 kN/m.$^{2.}$ Ground roughness is Class A ; basic snow load is $0.25\ \mathrm{kN/m^2}$, corrected foundation

Bearing capacity characteristic value $f_a=200\mathrm{kN/m^2}$Live load and Crane load combination coefficient ψ_c = 0.7Wind load combination value system

The value is taken as 0.6 . The quasi-permanent value coefficient for crane load is $\psi_{.q}$ □ 0.6The quasi-permanent coefficients for both live load and wind load are $\psi_{.q}$ □ 0.

A frame structure design is required (seismic fortification not considered)

2 Structural component selection and cross-sectional dimension determination

Because the span of the factory building is between 15 and 36 meters , and the column top elevation is greater than 8 meters, a reinforced concrete frame structure is adopted. To ensure greater roof rigidity, prestressed concrete polygonal roof trusses and prestressed concrete roof panels are selected. Reinforced concrete crane beams and foundation beams are also selected. The selection of the main components of the factory building is shown in Table 1.

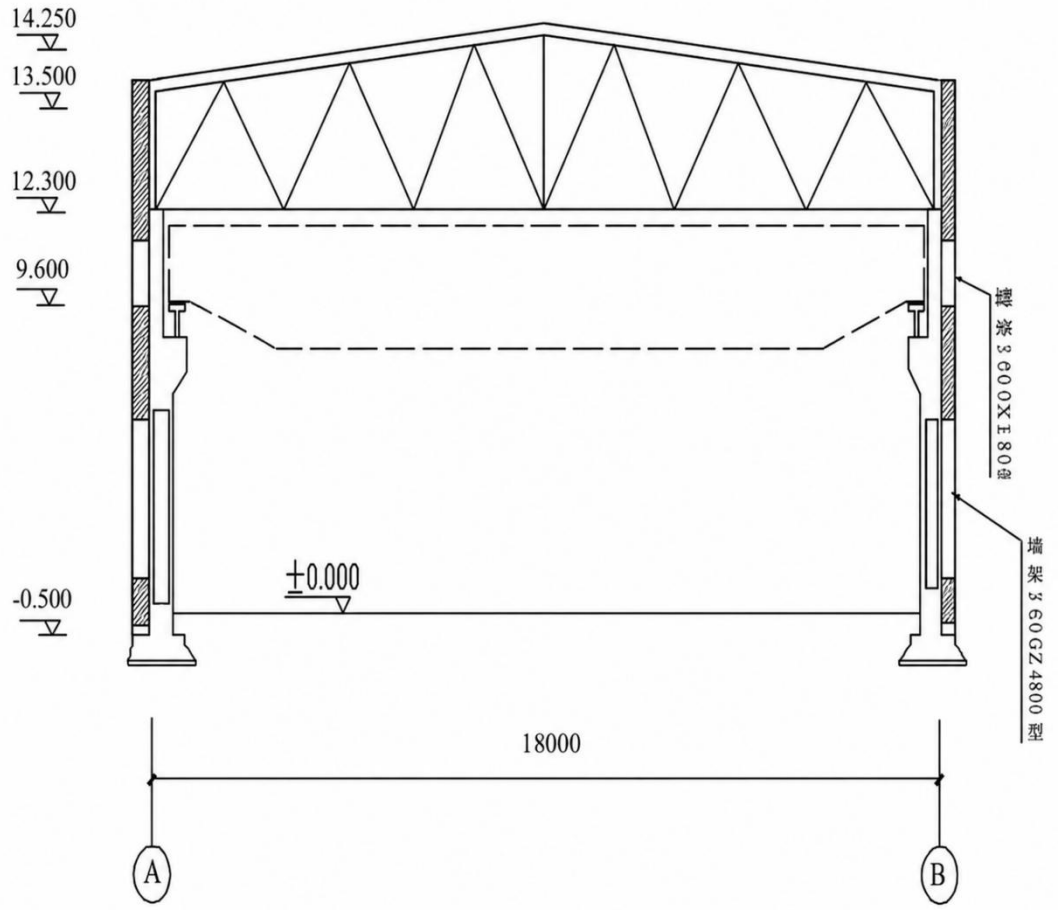


表 1 主要承重构件选型表

构件名称	标准图集	选用型号	重力荷载标准值
屋面板	G410（一） 1.5m×6m 预应力混凝土屋面板	YWB-3 II	1.40kN/m^2 (包括灌缝重)
屋架	G415（三） 预应力混凝土折线形屋架（跨度 24m）	YWJA-24-2C	109 kN/榀 0.05kN/m^2 （包括屋盖支撑系统）
吊车梁	G425 先张法预应力钢筋混凝土吊车梁 （吊车工作级别 A5）	YXDL6-6	44.2kN/根
轨道连接	G325 吊车轨道连接详图	DGL-14	0.808kN/根
基础梁	G320 钢筋混凝土基础梁	JL-22	23.4kN/根

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由图 1 可知柱顶标高为 12.6m，牛腿顶面标高为 8.7m，设室内地面至基础顶面的距离为 0.5m，则计算简图中柱的总高度 H，下柱 H_l 和上柱高度 H_u 分别为：

$H = 12.3m + 0.5m = 12.8m$　　$H_l = 8.7m + 0.5m = 9.2m$

$H_u = 12.8m - 9.2m = 3.6m$

根据柱的高度，吊车起重量及工作级别等条件，可确定柱截面尺寸，见表 2。

表 2 柱截面尺寸及相应的计算参数

柱号	计算参数	截面尺寸/mm	面积/mm^2	惯性矩/mm^4	自重/（kN/m）
A,B	上柱	矩 400×400	1.6×10^5	21.3×10^8	4.0
	下柱	I 400×900×100×150	1.875×10^5	195.138×10^8	4.69

选取 1 榀排架进行计算，计算单元和计算简图如图 2 所示。

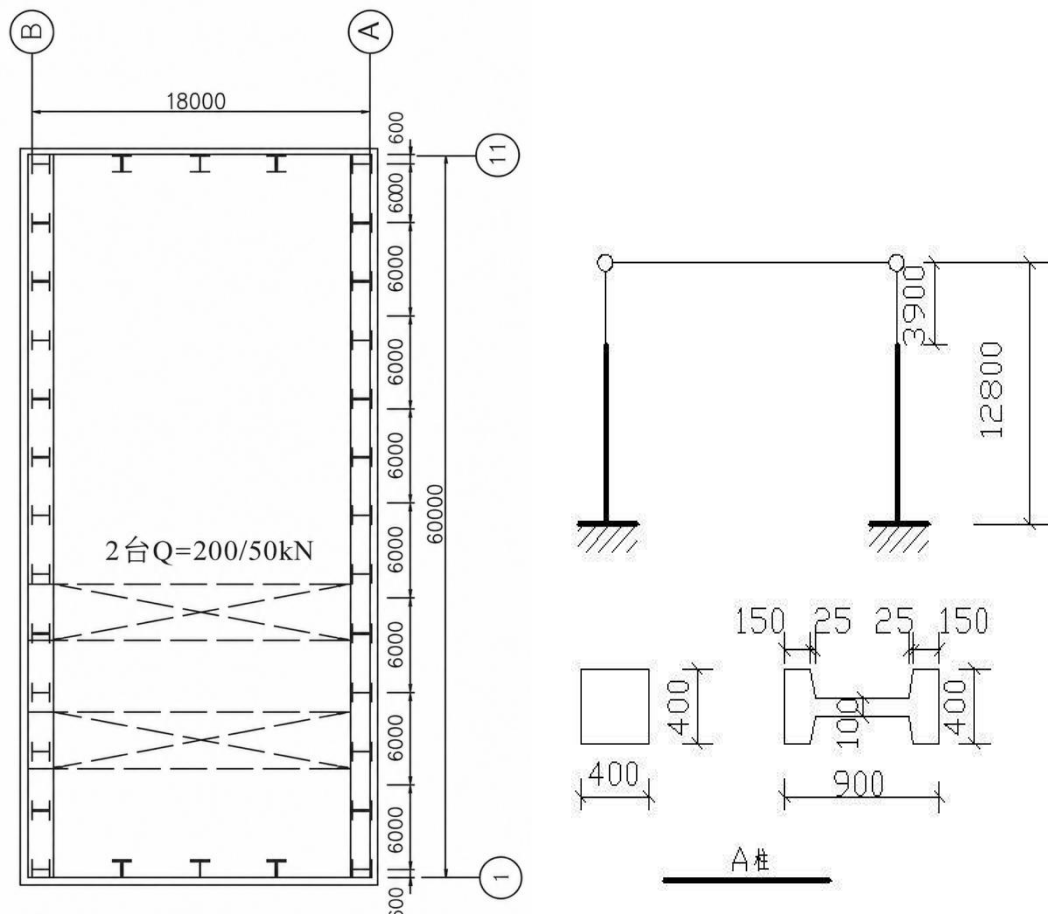


图 2 计算单元和计算简图

1 荷载计算

1.1 恒载

（1） 屋盖恒载

三毡四油防水层，绿豆砂护面	0.40kN/m^2
20mm 厚水泥砂浆找平层	20kN/m^3×0.02m=0.40kN/m^2
100mm 厚珍珠岩砼保温层	11kN/m^3×0.1m=1.10kN/m^2
20mm 厚水泥砂浆找平层	20kN/m^3×0.02m=0.40kN/m^2
屋面板自重（包括灌缝）	1.40kN/m^2
合计	3.70kN/m^2

屋架重力荷载为 109kN/榀，则作用于柱顶的屋盖结构重力荷载设计值

$$G_1 = 1.3 \times (3.7\text{kN/m}^2 \times 6\text{m} \times 24\text{m}/2 + 109\text{kN}/2) = 488.02\text{kN}$$

（2） 吊车梁及轨道重力荷载设计值

$$G_3 = 1.3 \times (44.2\text{kN} + 0.808\text{kN/m} \times 6\text{m}) = 63.763\text{kN}$$

（3） 柱自重重力荷载设计值

上柱 $G_{4A} = G_{4B} = 1.3 \times 4.0\text{kN/m} \times 3.6\text{m} = 18.72\text{kN}$

下　　柱 G_{5A}

$= G_{5B}$	=	× 9.2m	=
	1.3		56.09
	×		kN
	4.69		
	kN/m		

各项恒载作用位置如图 3 所示

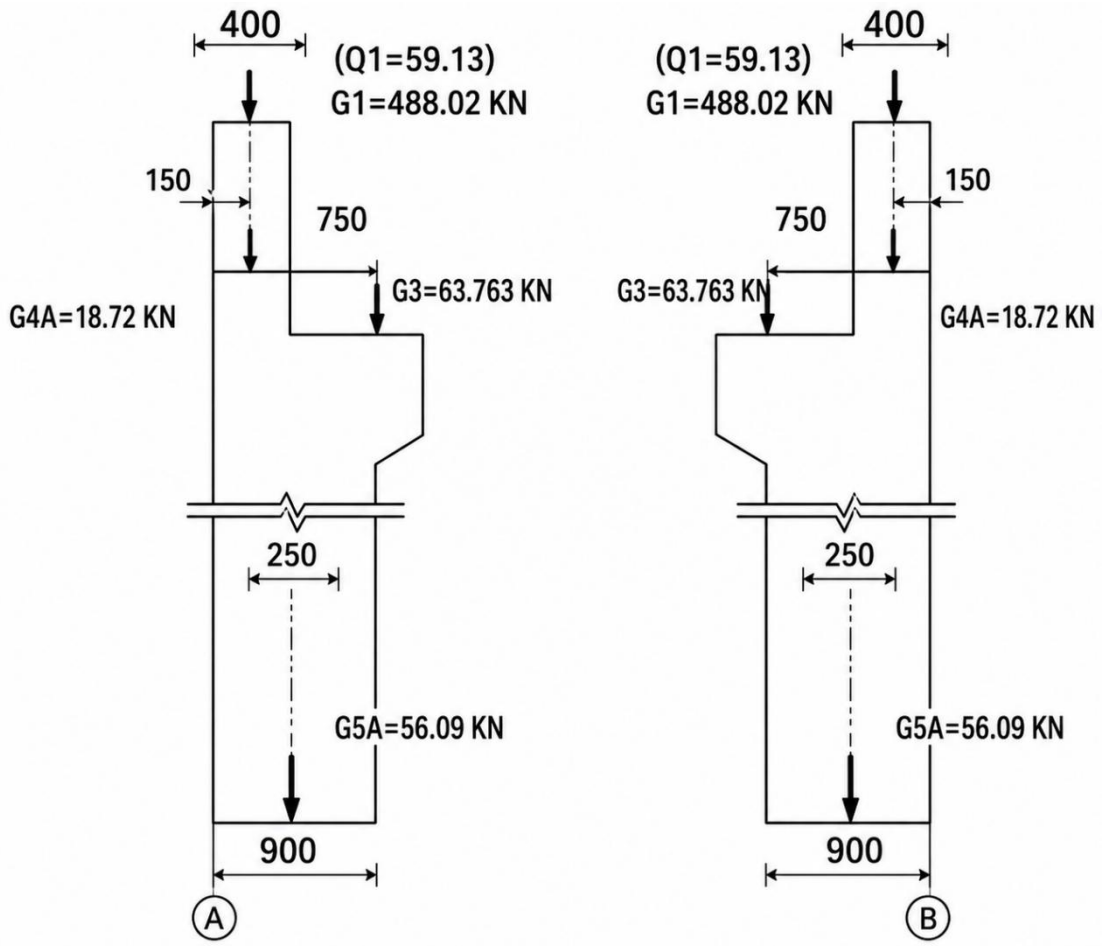


图 3 荷载作用位置图（单位：kN）

1.2 屋面活荷载

屋面活荷载标准值为 0.730kN/m^2，雪荷载标准值为 0.4kN/m^2，后者小于前者，故仅按前者计算，作用于柱顶的屋面活荷载设计值为：

$$Q_1 = 1.5\times 0.730\text{kN/m}^2 \times 6\text{m} \times 18\text{m}/2 = 59.13\text{kN}$$

Q_1的作用位置与G_1作用位置相同，如图 3 所示。

1.3 风荷载

风荷载标准值按式$\omega_k = \beta_z \mu_s \mu_z \omega_0$计算，其中$\omega_0 = 0.40\text{kN/m}^2$，$\beta_z = 1.0$

μ_z根据厂房各部分标高（图 1）及 A 类地面粗糙度由《混凝土结构设计》附表 5.1 确定如下：

柱顶（12.30m+0.15m=12.45m） $\mu_z = 1.449$

檐口（13.50m+0.15m=13.65m） $\mu_z = 1.4822$

屋顶（14.25m+0.15m=14.4m） $\mu_z = 1.5032$

μ_s如图 4a 所示，由式$w_k = \beta_z \mu_s \mu_z \omega_0$可得排架迎风面及背风面的风荷载标准值分别为：

$$w_{1k} = 1.0 \times 0.8 \times 1.449 \times 0.4\text{kN/m}^2 = 0.4637\text{kN/m}^2$$

w_{2k} $= 1.0 \times 0.5 \times 1.449 \times 0.4\text{kN/m}^2 = 0.2898\text{kN/m}^2$

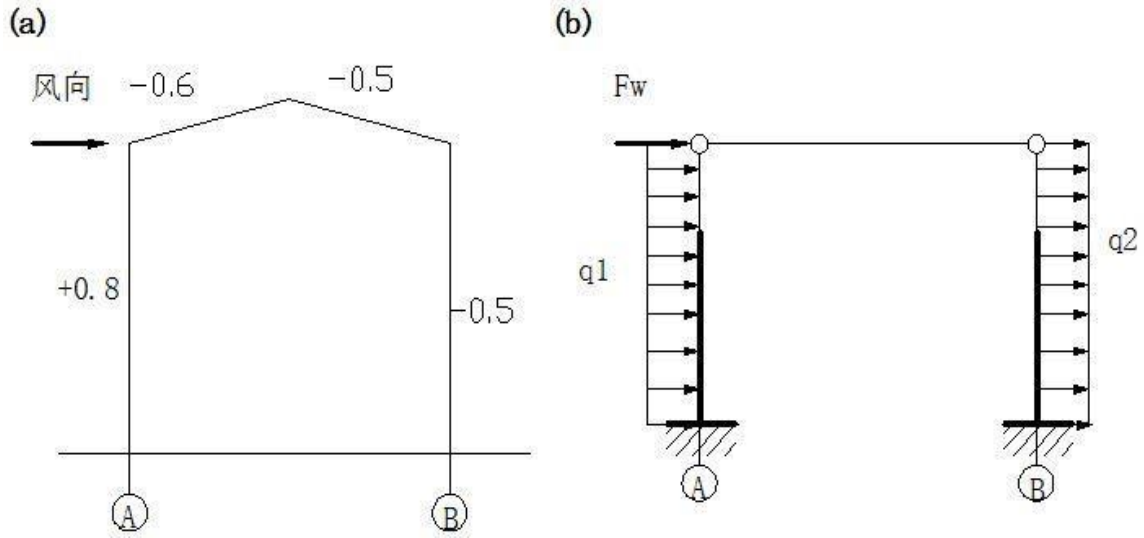


图 4 风荷载体型系数及排架计算简图

作用于排架计算简图 4b 上的风荷载设计值为：

$q_1 = 1.5 \times 0.4637\text{kN/m}^2 \times 6.0\text{m} = 4.1733\text{kN/m}$

$q_2 = 1.5 \times 0.2898\text{kN/m}^2 \times 6.0\text{m} = 2.6082\text{kN/m}$

$$F_w = \gamma_Q\left[\left(\mu_{s1} + \mu_{s2}\right)\mu_z h_1 + \left(\mu_{s3} + \mu_{s4}\right)\mu_z h_2\right]\beta_z\omega_0 B$$

$$= 1.5 \times \left[\left(0.8 + 0.5\right) \times 1.4822 \times 1.2\text{m} + (-0.6 + 0.5) \times 1.5032 \times .75\text{m}\right]$$

$$\times 1.0 \times 0.4\text{kN/m}^2 \times 6.0\text{m}$$

$$= 7.920\text{kN}$$

1.4 吊车荷载

由《混凝土结构设计》表 2.5.1 可得 200/50kN 吊车的参数为：

$B = 5.55\text{m}, K = 4.40\text{m}, g = 78\text{kN}, Q = 200\text{kN}, F_{p,\max} = 215\text{kN}, F_{p,\min} = 45\text{kN}$

根据 B 及 K，可算得吊车梁支座反力影响线中各轮压对应点的竖向坐标值，如图 5 所示。

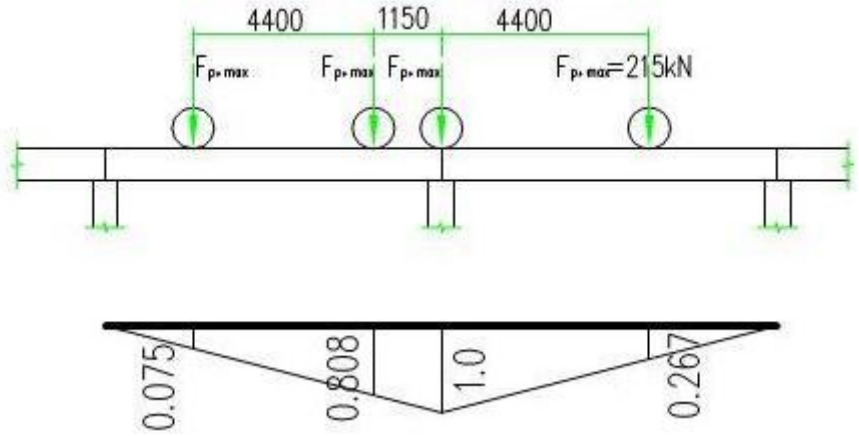


图 5 吊车荷载下支座反力影响线

（1） 吊车竖向荷载

由公式$D_{max} = \sum F_{pi,max} y_i$　　$D_{min} = \sum F_{pi,min} y_i$ 可得吊车竖向荷载设计值为：

$$D_{max} = \gamma_Q F_{p,max} \sum y_i = 1.5 \times 215\text{kN} \times (1.0 + 0.808 + 0.267 + 0.075) = 693.375\text{kN}$$

$$D_{min} = \gamma_Q F_{p,min} \sum y_i = 1.5 \times 45\text{kN} \times (1.0 + 0.808 + 0.267 + 0.075) = 145.125\text{kN}$$

（2） 吊车横向水平荷载

作用于每一个轮子上的吊车横向水平制动力按公式$T = \frac{1}{4}\alpha(Q + g)$计算

$$T = \frac{1}{4}\alpha(Q + g) = \frac{1}{4} \times 0.1 \times (200\text{kN} + 78\text{kN}) = 6.95\text{kN}$$

作用于排架柱上的吊车横向水平荷载设计值

$$T_{max} = \gamma_Q T \sum y_i = 1.5 \times 6.95\text{kN} \times 2.15 = 22.413\text{kN}$$

2 排架内力分析

该厂房为单跨排架，可用剪力分配法进行排架内力分析，其中柱的剪力分配系数η_i 按公式$\eta_i = \frac{1/\delta_i}{\sum_1^n 1/\delta_i}$ 计算，结果见表 3。

表 3 柱剪力分配系数

柱别	$n=I_u/I_l$ $\lambda=H_u/H$	$C_0=3/\left[1+\lambda^3(1/n-1)\right]$ $\delta=H^3/C_0EI_l$	$\eta_i=\frac{1/\delta_i}{\Sigma 1/\delta_i}$
A,B 柱	$n = 0.109$ $\lambda = 0.281$	$C_0 = 3.06$ $\delta_A=\delta_B=.208\times 10_^{10}\times\frac{H^3}{E}$	$\eta_A=\eta_B=0.5$

2.1 恒载作用下排架内力分析

恒载作用下排架的计算简图如图 6a 所示。图中的重力荷载$\overline{G}$及力矩 M 是根据图 3 确定的，即：

$\overline{G_1} = G_1 = 488.02kN$ ；$\overline{G} = G_3 + G_{4A} = 63.76\text{kN} + 18.72\text{kN} = 82.48\text{kN}$

$\overline{G_3} = G_{5A} = 63.763\text{kN}$

$M_1 = \overline{G_1}e_1 = 488.02\text{kN}\times 0.05\text{m} = 24.401\text{kN}\cdot\text{m}$

$M_2 = \left(\overline{G_1} + G_{4A}\right)e_0 - G_3e_3 = (488.02+18.72)\times 0.25 - 63.763\times 0.3 = 107.5561kN\cdot m$

由于图 6a 所示排架结构无侧移，故各柱可按柱顶为不动铰支座计算内力。柱顶不动铰支座反力 R_i 可根据《混凝土结构设计》表 2.5.2 所列的相应公式计算。对于 A,B 柱，$n = 0.109$ $\lambda = 0.281$，则

$$C_1=\frac{3}{2}\times\frac{1-\lambda^2(1-\frac{1}{n})}{1+\lambda^3(\frac{1}{n}-1)}=2.53 \qquad C_3=\frac{3}{2}\times\frac{1-\lambda^2}{1+\lambda^3(\frac{1}{n}-1)}=1.42$$

$R_A=\frac{M_1}{H}C_1+\frac{M_2}{H}C_3 = \frac{24.401\times 2.53+107.5561\times 1.42}{12.8}=16.755\ KN\ (\rightarrow)$

$R_B = -16.755$ kN（$\leftarrow$）

求得 R_i 后，可用平衡条件求出柱各截面的弯矩和剪力。柱各截面的轴力为该截面以上重力荷载之和，恒载作用下排架结构的弯矩图和轴力图分别见图 6b,c。

图 6d 为排架柱的弯矩、剪力和轴力的正负号规定。

2.2 屋面活荷载作用下排架内力分析

A,B 跨作用屋面活荷载，排架计算简图如图 7 所示。其中Q_1 = 73.584kN ，它在柱顶及变阶处引起的力矩为M_{1A} = 73.584kN × 0.05m = 3.68kN · m，

M_{2A} = 73.584kN × 0.25m = 18.40kN · m

对于 A,B 柱，C_1 = 2.128 , C_3 = 1.124 , 则

$$R_A = R_B = \frac{M_1}{H}C_1 + \frac{M_{2A}}{H}C_3 = \frac{3.68\text{kN} \cdot \text{m} \times 2.128 + 18.40\text{kN} \cdot \text{m} \times 1.124}{13.1\text{m}} = 2.176kN(\leftarrow)$$

R_B = -2.176kN(

2.3 风荷载作用下排架内力分析

（1） 左风吹时

计算简图如图 8.1 所示。对于 A,B 柱， n = 0.109， λ = 0.298，由《混凝土结构设计》表 2.5.2 所得：

$$C_{11} = \frac{3\left[1+\lambda^4\left(\frac{1}{n}-1\right)\right]}{8\left[1+\lambda^3\left(\frac{1}{n}-1\right)\right]} = 0.328$$

$R_A = -q_1 H C_{11}$ = −3.92kN/m × 13.1m × 0.328 = −16.84kN （←）

$R_B = -q_2 H C_{11}$ = −2.45kN/m × 13.1m × 0.328 = −10.53kN （←）

$R = R_A + R_B + F_w$ = −16.84kN − 10.53kN − 15.50kN = −41.87kN （←）

各项顶剪力分别为：

$V_A = R_A - \eta_A R$ = −16.84kN + 0.5 × 41.87kN = 4.10kN (→)

$V_B = R_B - \eta_B R$ = −10.53kN + 0.5 × 41.87kN = 10.40kN (→)

（2） 右风吹时

由于单跨，与左风吹情况相同，内力反向，计算简图和内力图如图 8.2

图 8.2 右吹风时排架内力图

2.4 吊车荷载作用下排架内力分析

（1） D_{max} 作用于 A 柱

计算简图如图 9a 所示，其中吊车竖向荷载 D_{max}，D_{min} 在牛腿顶面处引起的力矩为

$M_A = D_{max}e_3 = 647.15\text{kN} \times 0.3\text{m} = 194.15\text{kN} \cdot \text{m}$

$M_B = D_{min}e_3 = 135.45\text{kN} \times 0.3\text{m} = 40.64\text{kN} \cdot \text{m}$

对于 A 柱，$C_3 = 1.124$，则

$$R_A = -\frac{M_A}{H}C_3 = -\frac{194.15\text{kN} \cdot \text{m}}{13.1\text{m}} \times 1.124 = -16.66\text{kN} \ (\leftarrow)$$

对于 B 柱

$$R_B = \frac{M_B}{H}C_3 = \frac{40.64\text{kN} \cdot \text{m}}{13.1\text{m}} \times 1.124 = 3.49\text{kN} \ (\rightarrow)$$

$R = R_A + R_B = -16.66\text{kN} + 3.49\text{kN} = -13.17\text{kN}$ （←）

排架各柱剪力分别为

$V_A = R_A - \eta_A R = -16.66\text{kN} + 0.5 \times 13.17 = -10.08\text{kN}$ （←）

$V_B = R_B - \eta_B R = 3.49\text{kN} + 0.5 \times 13.17 = 10.08\text{kN} \ (\rightarrow)$

排架各柱的弯矩图、轴力图及柱底剪力值如图 9 所示。

（2） D_{max} 作用在 B 柱

计算简图如图 10a 所示，将“ D_{max} 作用在 A 柱”的情况的 A,B 柱内力对换，并注意改变符号，渴求的 D_{max} 作用于 B 柱时各柱的内力。排架各柱的弯矩图、轴力图及柱底剪力值如图 10b,c 所示。

（3）T_{max} 作用于 AB 跨柱

当 AB 跨作用吊车横向水平荷载时，排架计算简图如图 11a 所示。对于 A 柱，$n = 0.109$，$\lambda = 0.298$，由《混凝土结构设计》表 2.5.3 得

$\alpha = (3.9 - 1.2)/3.9 = 0.692$，则

$$C_5 = \frac{2 - 3\alpha\lambda + \lambda^3\left[\frac{(2+\alpha)(1-\alpha)^2}{n} - \left(2 - 3\alpha\right)\right]}{2\left[1 + \lambda^3\left(\frac{1}{n} - 1\right)\right]} = 0.594$$

$R_A = R_B = -T_{max}C_5 = -20.92\text{kN} \times 0.594 = -12.43\text{kN}$ （←）

排架柱顶总反力 R 为：$R = R_A + R_B = -12.43kN - 12.43kN = -24.86kN$

考虑厂房整体空间作用：$\mu = 0.85$

$V_A = V_B = R_A - \eta\mu R = -12.43\text{kN} + 0.5 \times 0.85 \times 24.86\text{kN} = -1.86\text{kN}$ （←）

排架各柱的弯矩图及柱底剪力值如图 11b 所示。当$T_{\max}$方向相反时弯矩图和剪力只改变符号，方向不变。

只考虑风荷载时 A 柱的内力组合　　表 4-14

截面		内力	$+M_{max}$ 及相应的 N、V		$-M_{max}$ 及相应的 N、V		N_{max} 及相应的 M、V		N_{min} 及相应的 M、V	
			组合项	数值	组合项	数值	组合项	数值	组合项	数值
A柱	Ⅰ—Ⅰ	M (kN·m)	1+0.9×(2+5+10×11)	35.73	1+0.9×[(4+6)×0.8+9+12]	−40.58	1+0.9×[(4+6)×0.8+0.9+12]	−40.58	1+0.9×[(4+6)×0.8+9+12]	−40.58
		N (kN)		223.86		189.84		189.84		189.84
	Ⅱ—Ⅱ	M (kN·m)	1+0.9×[0.8×(3+5)+10+11]	68.55	1+0.9×(2+6+9+12)	−29.41	1+0.9×[2+(3+5)×0.8+10+11]	64.42	1+0.9×(5+10+11)	24.48
		N (kN)		445.03		276.42		479.05		242.4
	Ⅲ—Ⅲ	M (kN·m)	1+0.9×(2+5+10+11)	241.34	1+0.9×[0.8×(4+6)+7+12]	−197.95	1+0.9×(2+3+8+11)	187.64	1+0.9×(5+10+11)	234.87
		N (kN)		364.74		401.26		618.03		330.72
		V (kN)		8.28		−4.75		−6.16		7.13

Load Case	Section I-I (M/N/V)	Section II-II (M/N/V)	Section III-III (M/N/V)
1. Dead Load (D)	19.25 / 385.08 / 0	20.82 / 403.80 / 0.44	-32.04 / 514.44 / -5.74
2. Roof Live Load (L_r)	2.91 / 58.12 / 0	3.15 / 60.94 / 0.07	-4.84 / 77.63 / -0.87
3. Left Wind (W_L)	0 / 0 / 4.10	14.76 / 0 / 4.10	52.48 / 0 / 4.10
4. Right Wind (W_R)	0 / 0 / -4.10	-14.76 / 0 / -4.10	-52.48 / 0 / -4.10
5. Crane D_{max}on A (C_{A1})	0 / 0 / -10.07	-36.25 / 647.15 / -10.07	-128.91 / 647.15 / -10.07
6. Crane D_{max} on B (C_{A2})	0 / 0 / 10.07	36.25 / 135.45 / 10.07	128.91 / 135.45 / 10.07
7. Crane Horiz. Tmax (→)(T_L)	0 / 0 / -1.86	-6.70 / 0 / -1.86	-23.81 / 0 / -1.86
8. Crane Horiz. Tmax(←)(T_R)	0 / 0 / 1.86	6.70 / 0 / 1.86	23.81 / 0 / 1.86

Control Section	Governing Combination Case	Design Moment M (kN·m)	Design Axial Force N (kN)	Design Shear Force V (kN)
Section I-I	**+M (max) & N (max)**	**29.39**	**587.78**	**0.00**

Control Section	Governing Combination Case	Design Moment M (kN·m)	Design Axial Force N (kN)	Design Shear Force V (kN)
(Top of Upper Column)	(1.3D + 1.5L_r)			
Section II-II *(Bottom of Upper Column)*	**一、 +M (max)** (1.3D + 1.35[L_r+ C_{A2} + W_L])	**88.35**	**741.45**	**19.63**
	二、 -M (max) (N (max)) (1.3D + 1.35[L_r + (C_{A1}) + (W_R)])	**-37.59**	**1432.24**	**-18.47**
Section III-III *(Base of Lower Column)*	**一、 +M (max)** (1.3D + 1.275[[L_r + C_{A2} + W_L + T_R])	**192.83**	**839.81**	**11.45**
	二、 -M (max) (N (max)) (1.3D + 1.275[[L_r + C_{A1} + W_R + T_L)	**-299.41**	**1492.23**	**-28.79**
	三、N (min) (1.3D + 1.5W_R)	**-120.37**	**668.77**	**-11.89**

Control Section	Combination Type	Bending Moment M (kN·m)	Axial Force N (kN)	Shear Force V (kN)	Load Combination Case
Section I-I	+Mmax (Nmax)	24.40	385.08	0.00	1.3 × Dead Load + 1.5 × Roof Live Load
Section II-II	+Mmax	107.56	403.80	14.76	1.3 × Dead Load + 1.5 × (Left Wind + Crane Dmax on Column B)
Section II-II	−Mmax	−36.25	647.15	−10.07	1.3 × Dead Load + 1.5 × (Right Wind +

Control Section	Combination Type	Bending Moment M (kN·m)	Axial Force N (kN)	Shear Force V (kN)	Load Combination Case
					Crane Dmax on Column A)
Section II-II	Nmax	−36.25	1050.95	−10.07	1.3 × Dead Load + 1.5 × Crane Dmax on Column A
Section III-III	+Mmax	213.97	514.44	11.59	1.3 × Dead Load + 1.5 × (Left Wind + Crane Dmax on Column B + Tmax to the Right)
Section III-III	−Mmax	−299.41	1161.59	−28.79	1.3 × Dead Load + 1.5 × (Right Wind + Crane Dmax on Column A + Tmax to the Left)
Section III-III	Nmax	−128.91	1161.59	−10.07	1.3 × Dead Load + 1.5 × Crane Dmax on Column A
Section III-III	Nmin	−32.04	514.44	−5.74	1.3 × Dead Load (Considering minimum axial force for foundation uplift and sliding resistance check)

Sect6. Column Section Design

6.1 Material Properties and Design Parameters

Concrete (C30)

Axial compressive strength design value:$f_c = 14.3\ \text{N/mm}^2$

Modulus of elasticity:$E_c = 3.0 \times 10^4\ \text{N/mm}^2$

Reinforcement Steel (HRB400)

Tensile and compressive design strength:$f_y = f_y' = 360\ \text{N/mm}^2$

Modulus of elasticity:$E_s = 2.0 \times 10^5\ \text{N/mm}^2$

Section Dimensions (Upper Column)

Rectangular cross-section:

$b \times h = 400\ mm \times 400\ mm$

Concrete cover thickness:$c = 30\ mm$

Assuming rebar diameter $d = 20\ mm$ and stirrup diameter $dv = 8\ mm$, the effective distance is:

$as = as' = c + dv + d/2 = 30 + 8 + 10 = 48\ mm$

For calculation convenience:

$as = as' = 45\ mm$ Effective height:$h0 = h - as = 400 - 45 = 355\ mm$

Cross-sectional area:$A = 400 \times 400 = 1.6 \times 10^5\ mm^2$

6.2 Structural Slenderness and Length Effects

The unsupported length of the upper column in the transverse bent frame direction is taken as:$l0 = 2Hu$

From the structural model: $Hu = 3.6\ m = 3600\ mm$ Therefore:

$l0 = 2 \times 3600 = 7200\ mm$

Checking slenderness ratio: $l0 / h = 7200 / 400 = 18 > 8$

Since $l0/h > 8$, the structural second-order effects (η-effect) due to column slenderness must be considered by amplifying the initial eccentricity.

6.3 Calculation of Reinforcement for Section I–I (Upper Column Top)

From the force combination matrix, the governing design internal forces under the load combination:

1.3 × Dead Load + 1.5 × Roof Live Load are:

$M = 24.40\ kN \cdot m$ $N = 385.08\ kN$

1. Initial Eccentricity (e0)

$e0 = M / N = (24.40 \times 10^6) / (385.08 \times 10^3) = 63.36\ mm$

2. Additional Eccentricity (ea)

According to the design code: $h / 30 = 400 / 30 = 13.33$ mm

Compare with 20 mm: $ea = 20$ mm

Total initial eccentricity: $ei = e0 + ea = 63.36 + 20 = 83.36$ mm

3. Eccentricity Magnification Factor (η)

$\eta = 1 + [1 / (1300 \times (ei / h))] \times (l0 / h)^2 \times \zeta1 \times \zeta2$

Where: $\zeta1 = (0.5 \times fc \times A) / N$

$= (0.5 \times 14.3 \times 1.6 \times 10^5) / (385.08 \times 10^3) = 2.97$

Since $\zeta1 > 1.0$: $\zeta1 = 1.0$

For rectangular sections:

$\zeta2 = 1.0$ Therefore:

$\eta = 1 + [1 / (1300 \times (83.36 / 400))] \times (18)^2 \times 1.0 \times 1.0 = 1 + 1.196 = 2.20$

Design amplified eccentricity: $e = \eta ei + h/2 - as$

$= 2.20 \times 83.36 + 400/2 - 45 = 183.39 + 200 - 45 = 338.39$ mm

4. Boundary Relative Concrete Compression Zone Depth (ξb)

For HRB400 reinforcement: $\xi b = 0.8 / [1 + fy / (Es \times \varepsilon cu)]$

$= 0.8 / [1 + 360 / (2.0 \times 10^5 \times 0.0033)] = 0.518$

5. Verification of Compression Type

Assuming large eccentric compression:$\xi = N / (fc \times b \times h0)$

$= (385.08 \times 10^3) / (14.3 \times 400 \times 355) = 0.189$

Since:$\xi = 0.189 < \xi b = 0.518$

the member is confirmed as a Large Eccentric Compression member.

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6. Reinforcement Area Calculation (As = As′)

Using symmetrical reinforcement formulation:

$As = As' = [Ne - fc \times b \times h0^2 \times \xi \times (1 - 0.5\xi)] / [fy \times (h0 - as')]$

First: $Ne = 385.08 \times 10^3 \times 338.39 = 1.303 \times 10^8$ N·mm

Concrete compression contribution:

$fc \times b \times h0^2 \times \xi \times (1 - 0.5\xi) = 14.3 \times 400 \times 355^2 \times 0.189 \times (1 - 0.5 \times 0.189) = 1.233 \times 10^8$ N·mm

Therefore:

$As = As' = (1.303 \times 10^8 - 1.233 \times 10^8) / [360 \times (355 - 45)] = (7.0 \times 10^6) / 111600$

$= 62.72$ mm²

---

7. Minimum Reinforcement Ratio Check

According to the code:

$\rho min = 0.2\%$

Minimum reinforcement area:

$As,min = 0.002 \times b \times h = 0.002 \times 400 \times 400 = 320$ mm² Since:

62.72 mm² < 320 mm² the design is governed by the minimum reinforcement requirement.

Therefore:

$As = As' = 320$ mm²

6.4 Calculation of Reinforcement for Section II–II (Upper Column Bottom)

From the force combination table, the governing load combination is:

1.3 × Dead Load + 1.5 × (Left Wind + Crane Dmax on Column B)

Design internal forces:

M = 107.56 kN·m
N = 403.80 kN

---

1. Initial Eccentricity (e0)

$e_0 = M / N = (107.56 \times 10^6) / (403.80 \times 10^3) = 266.37$ mm

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2. Additional Eccentricity (ea)

$e_a = 20$ mm $e_i = e_0 + e_a = 266.37 + 20 = 286.37$ mm

---

3. Eccentricity Magnification Factor (η)

$\zeta_1 = (0.5 \times f_c \times A) / N = (0.5 \times 14.3 \times 1.6 \times 10^5) / (403.80 \times 10^3) = 2.83$

Since $\zeta_1 > 1.0$: $\zeta_1 = 1.0$ $\zeta_2 = 1.0$ Therefore:

$\eta = 1 + [1 / (1300 \times (286.37 / 400))] \times (18)^2 \times 1.0 \times 1.0 = 1 + 0.348 = 1.35$

Design amplified eccentricity:

$e = \eta e_i + h/2 - a_s = 1.35 \times 286.37 + 200 - 45 = 386.60 + 155 = 541.60$ mm

---

4. Verification of Compression Type

$\xi = N / (f_c \times b \times h_0)$

$= (403.80 \times 10^3) / (14.3 \times 400 \times 355) = 0.199$

Since:

$\xi = 0.199 < \xi b = 0.518$

the member is a Large Eccentric Compression member.

---

5. Reinforcement Area Calculation (As = As′)

$Ne = 403.80 \times 10^3 \times 541.60$

$= 2.187 \times 10^8$ N·mm

Concrete compression contribution:

$14.3 \times 400 \times 355^2 \times 0.199 \times (1 - 0.5 \times 0.199) = 1.299 \times 10^8$ N·mm

Therefore:

$As = As' = (2.187 \times 10^8 - 1.299 \times 10^8) / [360 \times (355 - 45)] = (8.88 \times 10^7) / 111600$

$= 795.70$ mm²

---

6. Reinforcement Selection

Minimum required reinforcement area:

$As,min = 320$ mm²

Since: 795.70 mm² > 320 mm² the calculated reinforcement governs.

Select:

4Φ16 longitudinal reinforcement bars on each side. Provided reinforcement area:

$As = As' = 4 \times 201.1 = 804.4$ mm² > 795.70 mm² Total steel area:

$804.4 \times 2 = 1608.8$ mm²

Steel ratio:

$\rho = 1608.8 / (400 \times 400) = 1.01\%$

This satisfies the code requirement: $0.55\% \leq \rho \leq 3\%$

6.6 Column Stirrup Configuration

From the force combination table, the maximum structural shear force at the column root occurs under Case A (Right Wind + Crane Action): Vmax = 28.79 kN

Checking the concrete shear capacity without shear reinforcement:

$0.7f_t bwh_0 = 0.7 \times 1.43\ N/mm^2 \times 100\ mm \times 850\ mm = 85.09\ kN$

Since: Vmax = 28.79 kN < 85.09 kN

the concrete section alone is sufficient to resist the design shear force, and no special shear reinforcement calculation is required.According to standard structural detailing requirements for industrial reinforced concrete columns, a closed rectangular stirrup arrangement is adopted.

Stirrup Configuration

Stirrup diameter: Φ8 plain round bars Cross-sectional area of one stirrup leg: Asv = 50.3 mm^2

Stirrup Spacing in Column Shaft

The allowable maximum spacing is determined by: $s \leq \min(15d, 400\ mm)$

$= 15 \times 16 = 240\ mm$

Therefore, the regular stirrup spacing is selected as: s = 200 mm

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Stirrup Spacing in Encrypted Zones

To improve confinement and local stress resistance, the stirrup spacing is reduced within the critical regions: Top confinement zone beneath the corbel: h = 900 mm

Bottom confinement zone near the foundation connection: h = 900 mm

Within these regions, the stirrup spacing is reduced to: s = 100 mm

This detailing arrangement satisfies the structural code requirements for local load transfer, crack control, and ductility enhancement.

Corbel Design (Column Bracket Design)

1.

7.1 Geometric Parameters and Sizing Verification

The corbel extends outward from the lower column to support the crane runway beam. Based on structural arrangement and support requirements, the design dimensions of the corbel are established as follows:

Width of corbel: $b = 400$ mm

(equal to the column width)

Total height at the column interface: $h = 800$ mm

Outer edge height: $h1 = 450$ mm $\geq h/2 = 400$ mm

This satisfies the structural requirement.

Distance from the vertical load center to the column connection interface (shear span): $a = 300$ mm

Concrete cover thickness:

$c = 35$ mm

Assuming a main reinforcement diameter of 20 mm, the effective height is:

$h0 = h - as = 800 - 45 = 755$ mm

Checking the shear span-to-depth ratio: $a / h0 = 300 / 755 = 0.397$

Since: $0.397 < 1.0$ the member is confirmed as a Short Corbel structure.

According to the design code, when $a/h0 < 0.3$, the value may be taken as 0.3. In this design, the actual calculated value of 0.397 is used.

7.2 Section Capacity Verification

To prevent localized concrete crushing and diagonal compression failure before reinforcement yielding, the section must satisfy the structural code ultimate shear limit condition:

$$Fv,d \leq [0.65 \times fc \times b \times h0] / [1 + 3.5 \times (a / h0)]$$

Substituting the material properties and geometric parameters:

$fc = 14.3$ N/mm²

Right-side calculation:

= [0.65 × 14.3 × 400 × 755] / [1 + 3.5 × 0.397] = 2807080 / (1 + 1.390)

= 2807080 / 2.390 = 1174510 N = 1174.51 kN

Checking against the design shear force:

Fv,d = 686.35 kN ≤ 1174.51 kN

Therefore, the corbel section dimensions are sufficient to resist diagonal compressive failure under ultimate crane loading conditions.

7.3 Main Tensile Loop Reinforcement Calculation (As)

The longitudinal tensile reinforcement located at the upper zone of the corbel is calculated using the truss analogy model, which resists both the bending moment caused by eccentric vertical loading and the direct horizontal crane braking force.

The required reinforcement area is:

As = (Fv,d × a) / (0.85 × fy × h0) + (Fh,d / fy)

Substituting the design loads and material properties:

fy = 360 N/mm²

As = [(686.35 × 10^3 × 300) / (0.85 × 360 × 755)] + (36.47 × 10^3 / 360)

= (2.059 × 10^8 / 231030) + 101.31= 891.23 + 101.31 = 992.54 mm²

Checking the minimum reinforcement ratio requirement for corbels:

ρmin = 0.2%

Minimum required reinforcement area:

As,min = 0.002 × b × h0 = 0.002 × 400 × 755 = 604 mm²

Since:

992.54 mm² > 604 mm²

the calculated force demand governs the reinforcement selection.

Provide:

4Φ20 longitudinal tensile reinforcement bars.

Provided reinforcement area:As,provided = 4 × 314.2 = 1256.8 mm² > 992.54 mm²

7.4 Horizontal Closed Stirrup Configuration

To control vertical crack propagation within the diagonal compression strut region, horizontal closed stirrups are distributed throughout the upper two-thirds of the corbel height.

The total required horizontal stirrup area is:

$A_{svh} \geq 0.5 \times A_s$

$= 0.5 \times 992.54$

$= 496.27 \text{ mm}^2$

Select Φ8 closed stirrups with two legs per layer:

$A_{sv1} = 2 \times 50.3$

$= 100.6 \text{ mm}^2$

Required number of stirrup layers:

Number of layers = 496.27 / 100.6

= 4.93

Adopt 5 layers.

Therefore, provide:

5Φ8 horizontal closed stirrup loops

distributed vertically at:

s = 100 mm

starting directly below the top longitudinal tensile reinforcement bars.

This arrangement satisfies the structural code requirement for stirrup spacing between 100 mm and 150 mm.